

BX-01 BEST · OPEN-CELL · CLOSED-CELL · LABORATORY STARTING FORMULAS

Open-Cell and Closed-Cell Foam

Formulas first. **BX-01** is the BazzLab card — the mix others do not publish. OC-01 and CC-01 stay as lab screens. TDS, catalyst, memo and patent notes follow so every number can be checked.

01 Best formula · BX-01 / BX-02

HFO-1233zd closed-cell on TB-307 / TEROL 250. Costs less than CREAPOL 326 / 1500. No Mannich. Unique to t

01a 50 g lab cup + checklist

50 g CC (BX-01 + HFO) and 50 g OC (BX-02, no HFO), each with a tick-off bench list.

02 Open-cell formula · OC-01 / OC-02

High-water memo screen. Soft, air-open, 7–12 kg/m³.

03 Closed-cell formula · CC-01 / CC-02

Lower-cost swap starting card. Sealed cells, 28–40 kg/m³. Mix about 90:100 with PMDI.

04 Polyol technical summary

ISOEXTER TB-307 / KIMpol PE 104 and TEROL 250 / KIMpol PE 103 in the cards, with client TDS numbers. PD 2 stays.

05 Catalysts and supplier memo ranges

PO 1009433 catalyst roles and xu wang systems A / B / C.

06 Mix method and trial matrix

Cup procedure, index maths, and the first pours — BX-01 first. 50 g weigh sheet is a separate page.

Best Formula · BX-01

BEST

NEW

This is the card to pour first. **HFO-1233zd(E) is mandatory** on closed cell. Memo System C and cup-test v4 are different recipes — do not blend them into this pour.

Why others do not have this

Typical spray copies use Mannich or sucrose for the network, CREAPOL 326 / 1500, and K-15 / DMCHA. They also use HFO — that part is now required here too.

BX-01 still does the rest differently. ISOEXTER TB-307 (f = 2.5) is the network — no Mannich. TEROL 250 (39.5 % aromatic) carries fire. PO 1009433 cats only. **HFO-1233zd(E) 10.0** is the blo gas. Water is cut to 1.20.

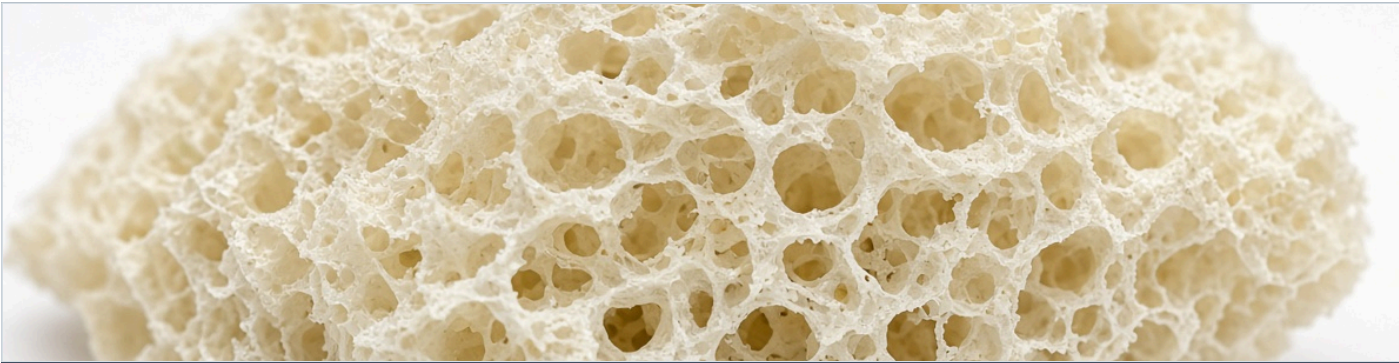
Target	
Density	32–38 kg/m³
Closed-cell content	> 90 %
Cream / gel / tack-free	3–6 s / 8–15 s / 15–28 s
Isocyanate	PMDI, 31.0 % NCO
Index	125 (calc.)
Weight ratio	≈ 83 A : 100 B

BX-01 · Closed-cell · HFO mandatory · 100.0 pbw B-side

Recommended first cup. Faster cream/gel pack. Confirm the drum label. Weigh HFO last, cold, then cap. Mix 4–5 s — cre now 3–6 s.

INGREDIENT	WHY IT IS IN THIS CARD — NOT IN THE USUAL COPIES	PBW	OH#
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost f = 2.5 rigid network. Replaces Mannich and CREAPOL 326.	41.3	305
TEROL 250 / KIMpol PE 103	Lower-cost flow + 39.5 % aromatic fire. Replaces CREAPOL 1500 and extra TCPP.	24.8	260
CREAPOL PD 230	Small toughness trim only. No lower-cost match.	4.0	225
TCPP	Cut vs typical 12–23. Terol aromatic does part of the fire job.	6.0	0
TEP	Viscosity cut for thick PE 104 / TB-307.	4.0	0
Reactive FR	Bound phosphorus. Re-index if the grade has OH.	2.0	0*
Emulsifier	Keeps 1.20 water in the thick polyester.	1.75	0
Closed-cell silicone	Sealed cells. Do not use an open-cell silicone.	1.5	0
Water	Cut because HFO does the blow. Do not go back to 1.75.	1.20	6233

Open-Cell Foam · OC-01



Open-cell structure. High water makes CO₂. Catalysts blow faster than they gel so the windows rupture.

Target		
Density		7–12 kg/m³
Closed-cell content		< 10 %
Cream / rise / tack-free		2–5 s / 8–15 s / 10–20 s
Isocyanate		PMDI, 31.0 % NCO
Index, lab cup		90–105
Index, 1:1 spray		35–55

OC-01 · 100.0 pbw B-side

Midpoints of supplier System C. Amine package is 10.0 pbw: blow (PC-37 + CAT-T + BDMAEE) over gel (PC-15).

INGREDIENT	WHY IT IS IN THE FORMULA
Sucrose / sucrose polyol	High-functionality backbone. Builds a network that can open and still stand.
Reactive flame retardant	FR that binds into the polymer instead of only sitting in the blend.
TEP	Cuts viscosity and adds phosphorus FR.
Ecosurf SA-9	Emulsifies water into the polyol so blow is even.
Silicone surfactant	Opens and evens the cells. Too little = collapse. Too much = closed windows.
Water	Main blowing agent. Reacts with NCO to make CO ₂ . This is what makes the foam open and light.
PC-37	Blow catalyst.
CAT-T	Blow catalyst.
PC-15	Gel / balance. Holds the rise so it does not collapse.
BDMAEE or DEDEE	Strong open-cell blow. Helps rupture windows.

Closed-Cell Foam · CC-01



Closed-cell structure. Low water, higher index, and gel-heavy catalyst keep the membranes intact.

Target	
Density	28–40 kg/m³
Closed-cell content	> 90 %
Cream / gel / tack-free	8–15 s / 25–45 s / 40–70 s
Isocyanate	PMDI, 31.0 % NCO
Index	115–125 (calc. 120)
Weight ratio	≈ 90 A : 100 B

CC-01 · 100.0 pbw B-side

Water is 2.0 pbw, not 15–26. That is the difference between a closed-cell rigid foam and an open-cell spray foam.

INGREDIENT	WHY IT IS IN THE FORMULA	PBW	OH#
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost high-OH rigid backbone. Replaces CREAPOL 326 GP.	35.5	305
TEROL 250 / KIMpol PE 103	Lower-cost flow polyester. Replaces CREAPOL 1500. Carries pentane in CC-02.	24.0	260
CREAPOL PD 230	Softer glycol polyester. Stops the blend going too brittle.	7.0	225
TCPP	Halogen FR.	12.0	0
TEP or DEHMP	FR plus viscosity cut.	6.5	0
Reactive FR	Bound FR. Re-index if it has OH groups.	5.0	0*
Emulsifier	Keeps water and FR mixed.	3.5	0
Silicone surfactant	Fine, sealed cells. Do not use an open-cell silicone here.	1.4	0
Water	Limited CO ₂ . Enough to help nucleation, not enough to open cells.	2.0	6233
PC-15	Gel / balance.	1.30	—

Polyol Technical Summary

CC-01 and OC-02 use TB-307 / TEROL 250, which cost less than CREAPOL 326 / 1500. Mid-range OH figures are used only for the CC-01 index. Specification windows stay as published.

PROPERTY	TB-307 / PE 104	TEROL 250 / PE 103	CREAPOL PD 230
Replaces	CREAPOL 326 GP	CREAPOL 1500	Kept — no lower match
Manufacturer NEW	COIM ISOEXTER TB-307 · Kimpur KIMpol PE 104	Huntsman TEROL 250 · Kimpur KIMpol PE 103	Flokser CREAPOL
Chemistry	Aromatic polyester, higher-OH rigid	Aromatic polyester for rigid PUR / PIR	Aromatic, glycol-polyester polyol
Stated use	Spray, systems, bunstock. Fire resistance plus mechanical properties.	Spray, pour-in-place, panels, rigid foam	General aromatic polyester modifier
Hydroxyl number	295–315 mg KOH/g (both)	PE 103: 250–270 TEROL 250: 235–255	210–240 mg KOH/g
Mid OH used in calc.	305 mg KOH/g	260 mg KOH/g (PE 103 mid)	225 mg KOH/g
Equivalent weight	183.9 g/eq @ 305	Card: 215.8 @ 260 TEROL published: 224.4	249.3 g/eq
Viscosity @ 25 °C	TB-307: 7 000 cP PE 104: 7 500–8 500 mPa·s	PE 103: 4 000–5 000 mPa·s TEROL: 4 000–6 000 cP	4250–4750 cP
Functionality NEW	TB-307: 2.5 · PE 104: not published	TEROL 250: 2.0 · PE 103: high-functionality	—
Density / SG NEW	TB-307: 1.20 g/cm³	TEROL 250: 1.234 @ 25 °C	—
Acid number NEW	TB-307: max 1.50 mg KOH/g	—	≤ 1 mg KOH/g
Aromatic content NEW	—	TEROL 250: 39.5 %	—
Storage / shelf NEW	Request official TDS	TEROL: 15–50 °C · 6 months	20–40 °C · 12 months
Role in the formulas	CC-01 rigid backbone	CC-01 flow / pentane carrier · OC-02	CC-01 modifier · trial

Client TDS pack — four grades, not two drums NEW

Catalysts and Supplier Ranges

Roles from PO 1009433. Chemical names are the industry spellings (the phone photos contain OCR typos). Quantities are listed only so the lab knows which drums are on the floor.

CODE	CHEMICAL NAME	ROLE IN FORMULA	ON THE PO	PACK	H
PC-15	N,N,N',N'-Tetramethyldipropylenetriamine	Gel / balance	26 236 lb	1 874 lb × 14 totes	2
PC-37	2-[2-(Dimethylamino)ethoxy]ethanol	Blow	1 984 lb	1 984 lb × 1 tote	2
CAT-T	Trimethyl hydroxyethyl ethylenediamine	Blow	9 370 lb	1 874 lb × 5 totes	2

Memo rule — most open cell	
Blow	CAT-T / BDMAEE / PC-37 / DEDEE
Gel / balance	PC-15
Strong gel / cure	DABCO 2040
Combined OC cat package	USD 5.15 / kg CIF Turkey port

Still required for the formulas
Polymeric MDI (≈ 31 % NCO), silicone surfactant, emulsifier (NP-9 or Ecosurf SA-9), flame retard (TCPP / TEP / DEHMP / reactive FR), biocide, and — for OC-01 — sucrose / polyether polyols. F every closed-cell cup, HFO-1233zd(E) is mandatory.

Supplier memo — OC/CC ingredient windows (pbw)

From Memo Style.pdf, 31 August 2026, xu wang to george@bazzfoam.com. These are ranges, not a finished 100-part for High water (15–26) marks A / B / C as open-cell type. Closed-cell water is 2.0 in CC-01.

INGREDIENT	A	B	C
Sucrose / sucrose polyol	20–60	—	30–36
Polyether polyol 1	—	15–19	—
Polyether polyol 2	—	14–18	—
Reactive FR	—	—	6–10
DEHMP / organophosphate	—	13–17	—
TEP	—	—	15–19
TCPP	15–25	—	—
Amine catalysts	4–10	8–12	8–12

Mix Method and Trial Matrix



Part B is the formula above. Part A is polymeric MDI. Bring both to temperature, then mix before cream.

Cup method

- 1. Condition A and B at **21–23 °C**.
- 2. Weigh B, then weigh A, into dry cups.
- 3. Pour A into B. Mix **4–5 s** at 2 500–3 000 rpm for both OC and CC — cream is now 2–6 s.
- 4. Clock cream, gel, rise and tack-free from mix start.
- 5. Do not touch after cream. Cut at 24 h.

First cup: 50.00 g B (includes 5.00 g HFO) + 41.40 g A on the [50 g lab](#) sheet. No machine spray until three clean cups.

Working equations

$$EW_{polyol} = 56\ 100 / OH\# \cdot eq = pbw / EW \cdot EW_{water} = 9.0 \cdot EW_{PMDI} = 4\ 200 / \%NCO$$
$$Index = (NCO\ eq / OH\ eq) \times 100 \cdot A\text{-side}\ (g) = OH\ eq \times (Index / 100) \times EW_{PMDI}$$

First pours — BX-01 first, then screens

TRIAL	CHANGE FROM CONTROL	WHAT YOU ARE ASKING
BX-01	Best card — HFO 10.0 · faster cat pack	Cream 3–6 s · gel 8–15 s · does it seal?
BX-01a	HFO 10.0 → 11.0 / water 1.20 → 1.10	Lighter / more blow
BX-01b	CAT-T +0.15 / PC-15 +0.15	Still faster if 3–6 / 8–15 is missed
BX-02	Best OC, only if sucrose is on the floor	Commercial 16–22 kg/m³ open cell without polyether
OC-01	Memo screen	Does System C open at 7–12 kg/m³?
CC-01	Lower-cost swap 90:100 screen	Fallback if BX-01 is too tight

Record

Temperature, mix time, cream, gel, rise, tack-free, collapse, shrink, scorch, friability.

Patent Notes and Sign-off

Numbers read from the client screenshots. Listed as blowing-agent / foam technology context, not freedom-to-operate advice.

APPLICANT (AS SHOWN)	PUBLICATION / PATENT	DATE ON SECOND SCREENSHOT
Honeywell	US7655610 / US7655610B2	2024.10
Honeywell	US8618040 / US8618040B2	2023.11
Honeywell	US8962707 / US8962707B2	2023.10
Honeywell	US9884946	2025.4.6
Honeywell	US8053404B2	2024.4
Honeywell	US8623808B2	2024.4
Arkema	CN105001440A	—
Honeywell	EP2640800A2 · EP2737006A2 · EP2606086A2 · EP3567072A1	—
Honeywell	JP5504161B2 · JP6231496B2	—

Source files used
ISOEXTER TB-307, KIMpol PE 103 / PE 104, TEROL 250 TDS summaries · TB307_PE104_TEROL250_PE103_TDS_Summary_Pack.pdf · CREAPOL 326 / 1500 / PD 230 T (reference) · Memo Style.pdf (xu wang, 31 Aug 2026) · PO 1009433 catalyst photos · Honeywell , Arkema patent screenshots.

What this pack is not
Not a production specification, fire-test certificate, spray-machine card, or legal clearance. Write the exact silicone, FR and PMDI grades on the trial sheet when the drums are opened.

Next actions

STEP	ACTION	OWN
1	Confirm sucrose / polyether, silicone, FR, PMDI, and which floor drum is open (TB-307 vs PE 104, TEROL 250 vs PE 103).	Purch
2	Run BX-01 first as a 200–300 g cup at 22 °C. Then OC-01 / CC-01 only as screens.	Lab
3	Change only one variable per trial.	Lab
4	Closed-cell cups must include HFO-1233zd(E). Hood on. Chill B. Weigh HFO last.	Lab